Madiha Khan

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SUMMARY

Software engineering student with experience building real-time monitoring dashboards and web applications using Python, JavaScript, React, and backend frameworks including FastAPI and Flask. Skilled in developing user-facing features, working with databases and APIs, and using Docker, Kubernetes, and Linux in collaborative engineering environments. Interested in scalable software and data-driven products.

EDUCATION

University of Calgary **Sep 2023 - Apr 2028**
Bachelor of Science, Software Engineering *Calgary, AB*
- **Coursework:** Data Structures & Algorithms, Full Stack Web Development, Applied Operating Systems, Software Testing & Architecture, Machine Learning, Networked Systems

EXPERIENCE

Chorate AI **Dec 2025 - Mar 2026**
Software Engineering Internship *Remote*
- Engineered real-time system monitoring dashboards by integrating OS-level metrics with a web frontend, applying performance optimization to proactively track resource utilization and server health.
- Contributed to assigned engineering tasks and feature development in a dynamic, remote team environment, supporting the delivery of scalable services and infrastructure.

Knowledge Bridge Engineering Club **May 2026 - Present**
Executive *Calgary, AB*
- Conducted educational research to align university course outcomes with functional, real-world tech industry expectations.

MISS UCalgary Club **May 2024 - Dec 2025**
VP of External Affairs *Calgary, AB*
- Managed corporate sponsorships and collaborated with cross-functional teams to fund and design large-scale professional networking events.

PROJECTS

LastSeen Room-memory Mobile App **May 2026**
- Developed a mobile application using Expo and React Native to capture room frames and process visual events through a custom change-detection data pipeline.
- Implemented a FastAPI API integrated with Supabase and an AI synthesis framework using OpenAI and Gemini to process natural-language queries for spatial item localization.
- Won 4th place at the Cursor Calgary Hackathon with a high-fidelity, demo-ready prototype featuring optimized image handling and a polished UX/UI layout.

Ice Cream Website **Jan 2026 - Apr 2026**
- Built a full-stack e-commerce ordering web application with custom REST APIs for user signup, secure login, persistent shopping carts, and past order histories.
- Implemented bulletproof authentication workflows using client-side session persistence, secure protected application routes, and backend bcrypt password hashing.
- Developed robust backend validation logic and reusable modular frontend components to support rapid product browsing, cart updates, and smooth checkout flows.

High-Throughput Ad Click Tracker **Aug 2026 - Aug 2026**
- Developed a scalable backend REST API using Python and FastAPI to ingest high-volume ad click events, leveraging Pydantic for strict data validation.
- Engineered a high-performance, in-memory caching layer using Redis transaction pipelines and atomic Lua scripts to efficiently aggregate time-bucketed event metrics.
- Implemented an asynchronous background worker that periodically drains cached Redis buckets to prevent system bottlenecks during traffic spikes.
- Designed a robust data pipeline utilizing to execute bulk operations into a PostgreSQL database, ensuring data integrity and preventing duplicate records on conflict.

TECHNICAL SKILLS

Languages: Python, C, C++, Java, JavaScript, SQL, HTML5/CSS3, MATLAB
Frameworks & Libraries: React.js, Next.js, React Native, Expo, Node.js, FastAPI, Flask, Pydantic, TailwindCSS
Developer Tools: Git, GitHub, Docker, Kubernetes, Linux/Unix Shell, VS Code, Cursor, RealTerm, MPLab, Continuous Integration/continuous Deployment (CI/CD)
Databases & Storage: MySQL, PostgreSQL, Supabase
Concepts & Methodologies: Object-Oriented Design, Data Structures & Algorithms, Embedded Systems, Debugging, Performance Optimization, UI/UX Implementation, API Development, Data Pipeline Design And Implementation, Scalable Service And Infrastructure Development